Imagine – An Exploratory Analysis

2026-03-10

Before we start

The data used in this report is the same you used for your analysis. I used the mean ratings for each dimension (factivity, intentionality, pictoriality) and scaled them. In the upcoming weeks, I will also adapt the scripts as to use the ratings of both coders to account for between-coder variance. I just didn't have the time to do everything. Since manual sense annotations are not yet available, I wrote the pipeline based on a clustering of the the data on the three provided annotation dimensions. We can later just replace them with the actual annotations.

Pre-analysis

Table 1.

Table 1: Pairwise Pearson Correlation

	intentionality	factivity	pictoriality
intentionality	1	-0.242	0.374
factivity		1.000	-0.222
pictoriality			1.000

Figure 1

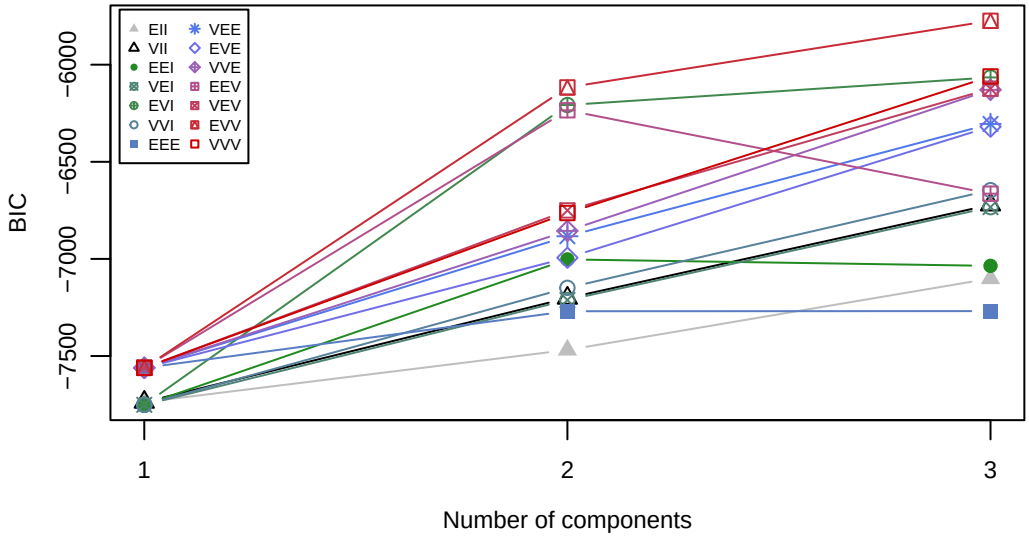


Figure 1: BIC for Gaussian mixture models.

Figure 2.

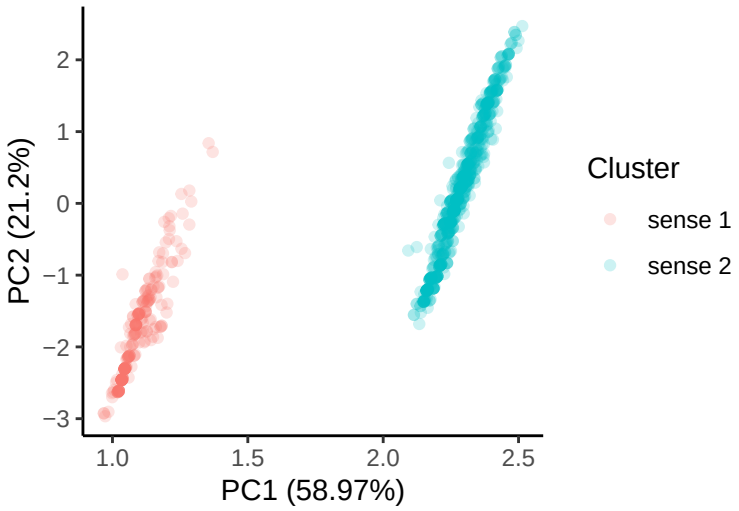


Figure 2: PCA of annotation space with latent sense clusters.

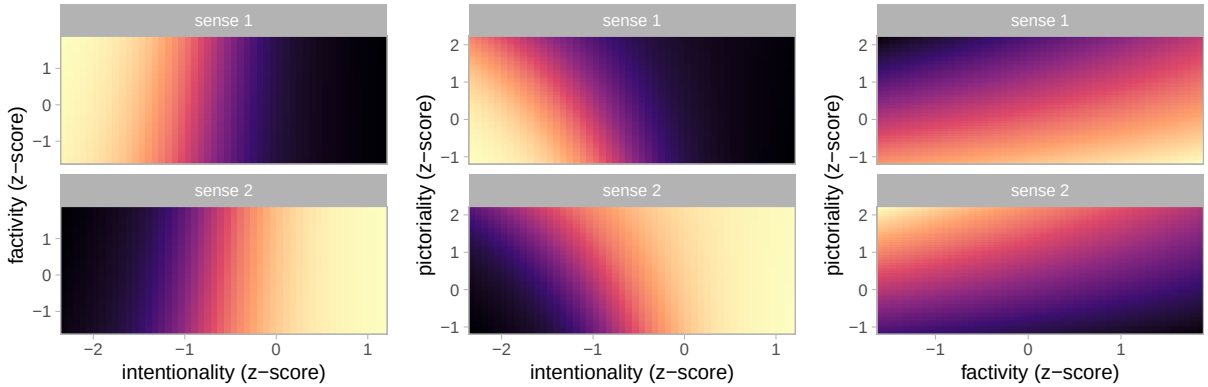


Figure 3: Predicted sense probability based on GLM with ridge regularization for 2-way interactions. The third dimension is held constant at the median value.The lighter the color, the higher the probability.

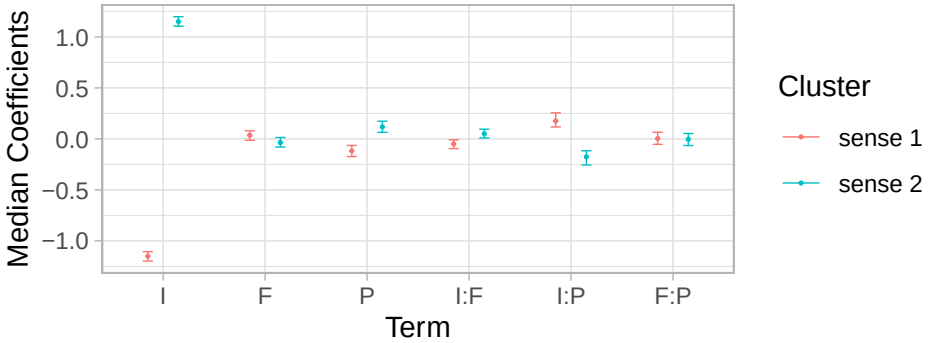


Figure 4: Bootstrapped Coefficients (n = 500) with 95% CI.

Embedding analysis

Centroids

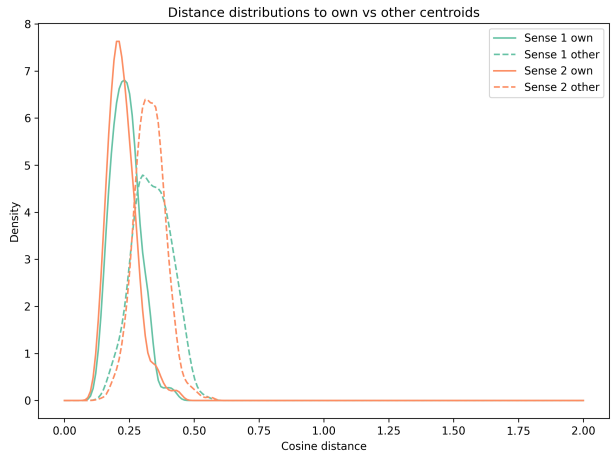
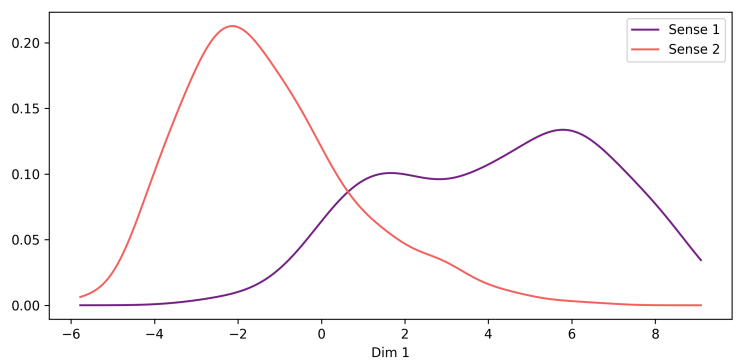
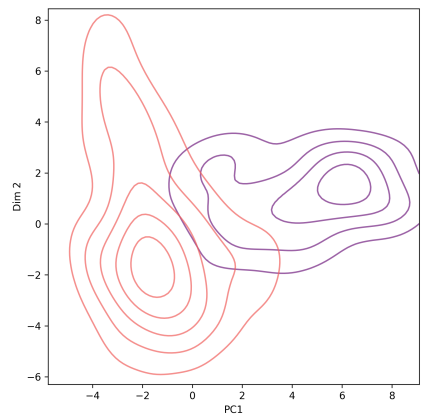


Figure 5: Caption



(a)



(b)

Embedding similarity vs sense similarity

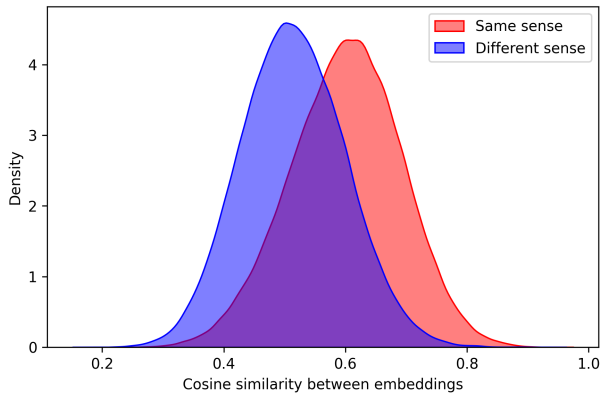


Figure 7: Caption

Senses under embedding projection

To examine how sentence embeddings capture the three annotated semantic dimensions (*intentionality*, *factivity*, and *pictoriality*), we trained simple linear regression models predicting each z-scaled dimension from the sentence embeddings. The regression coefficients define a direction in embedding space along which the dimension increases. By projecting each sentence embedding onto these directions, we obtain an embedding-based estimate for that dimension. This procedure was repeated for all three dimensions, resulting in a 3D projection of each sentence in the space spanned by the semantic dimensions.

Let $\mathbf{X} \in \mathbb{R}^{n \times d}$ denote the matrix of sentence embeddings, where n is the number of sentences and d the embedding dimension. Let $\mathbf{y}^{(k)} \in \mathbb{R}^n$ be the vector of z-scaled ratings for semantic dimension $k \in$ {intentionality, factivity, pictoriality}.

We fit a simple linear regression model for each dimension:

$$\mathbf{y}^{(k)} = \mathbf{X}\boldsymbol{\beta}^{(k)} + \boldsymbol{\epsilon}^{(k)},$$

where $\boldsymbol{\beta}^{(k)} \in \mathbb{R}^d$ is the vector of regression coefficients and $\boldsymbol{\epsilon}^{(k)}$ is the residual vector. The estimated coefficients $\hat{\boldsymbol{\beta}}^{(k)}$ define a direction in embedding space corresponding to increases in dimension k.

The projection of each sentence embedding $\mathbf{x}_i$ onto this direction provides an embedding-based estimate of the semantic dimension:

$$\hat{y}_i^{(k)} = \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}^{(k)}, \quad i = 1, \ldots, n.$$

By computing these projections for all three dimensions, each sentence is represented in a 3D space spanned by the semantic directions, allowing visualization and analysis of how embeddings capture the annotated semantic structure.

Figure 8 shows 3D projection of sentence embeddings onto the directions corresponding to the annotated semantic dimensions (intentionality, factivity, and pictoriality). It reveals that the sense clusters are well-separated in this space. This indicates that the latent embedding space encodes information aligned with the annotated dimensions. In other words, the directions learned via linear regression effectively capture the semantic structure underlying the sense distinctions. The clear separation suggests that even a simple linear mapping from embeddings to semantic dimensions is sufficient to discriminate the senses, highlighting that

the embeddings intrinsically encode patterns that correspond to the interpretive distinctions annotated in the dataset.

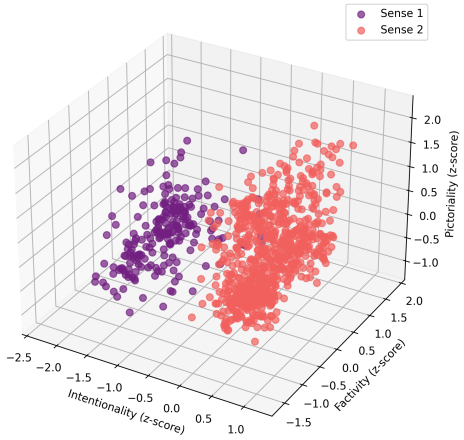
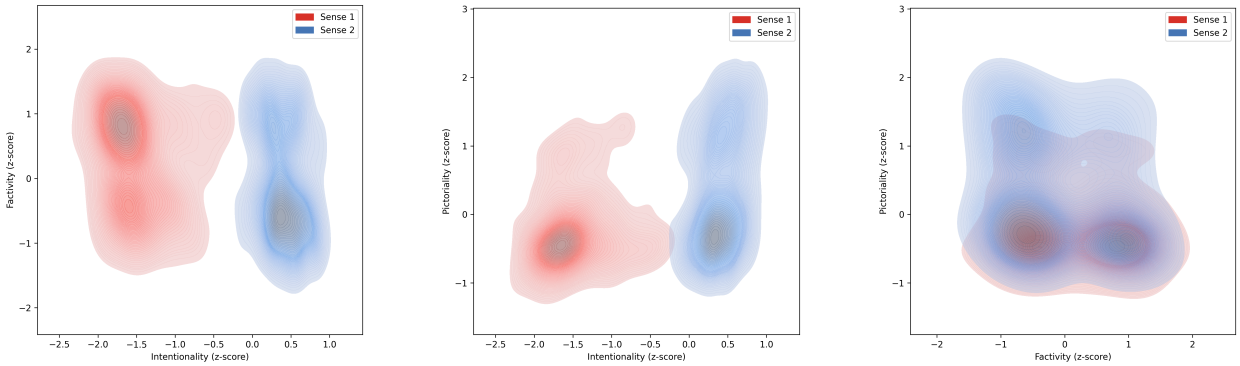


Figure 8: 3D embedding projections. Dimensions have been rescaled to z-scores.

Figure 9 depicts 2-dimensional for all combinations of dimensions. For projections for factivity and intentionality (Fig. 9a), sense 1 was predominantly associated with high factivity and low intentionality, whereas sense 2 showed the opposite pattern. Both distributions exhibited slight bimodality. In the pictoriality–intentionality projection (Fig. 9b), pictoriality contributed minimally, with sense clusters largely segregated along the intentionality axis. In contrast, the pictoriality–factivity projection (Fig. 9c) revealed substantial overlap between the two senses, indicating that the senses are not well distinguished along this dimension pair. These results suggest that the distinctions between the senses are primarily encoded along the intentionality and factivity dimensions, whereas pictoriality alone does not reliably differentiate sense membership.

Figure 9: 2D embedding projections. Dimensions have been rescaled to z-scores.



(a) (b) (c)